NICOLÁS GUARÍN GAVIRIA

Software & Cloud Data Engineer | DevOps & Backend | AWS · Azure · Python

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PROFESSIONAL SUMMARY

Software and Cloud data engineer with 4+ years building, deploying, and operating data pipelines, backend services, and ML workloads in banking and energy (Scotiabank, Bancolombia, ISA). Hands-on with Azure Databricks, Azure DevOps, and AWS (S3, Lambda, ECS Fargate, ECR, Bedrock, Transfer Family), and with the delivery side of the work: CI/CD pipelines, Docker, Terraform, and automated testing. Helped cut production deployment time at Bancolombia from 48 hours to 2. Outside work, I built and run an end-to-end data platform with scheduled GitHub Actions ingestion, a PySpark medallion lakehouse on Databricks and S3, and Terraform for an ECS Fargate deployment.

TECHNICAL SKILLS

Cloud (AWS): S3, Lambda, ECS Fargate, ECR, IAM, CloudWatch, Secrets Manager, DynamoDB, Bedrock, Transfer Family

Cloud (Azure): Azure Databricks, Azure Machine Learning, Azure DevOps (Repos, Pipelines, Boards)

DevOps & Deployment: CI/CD, Azure Pipelines, GitHub Actions (including self-hosted runners), Docker, Docker Compose, Terraform (Infrastructure as Code), Git, Kubernetes (fundamentals)

Data Engineering: ETL/ELT pipelines, PySpark, Databricks, Medallion architecture (bronze/silver/gold), DuckDB, Parquet, SQL, data quality checks, schema contracts, Power BI, SAS, Impala

Backend: Python, Java, FastAPI, REST APIs, microservices, PostgreSQL

ML & AI: XGBoost, scikit-learn, Azure ML, Retrieval-Augmented Generation (RAG), ChromaDB, LLM evaluation (Ragas)

Testing & Methods: Test-Driven Development, Playwright, Serenity BDD, pytest, Scrum, Agile, SDLC

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer **Apr 2025 – Present**

Scotiabank | Medellín, Colombia

- Introduced AI-assisted automation into the development lifecycle, shortening sprint cycle time from 7 days to 2–4 days.
- Contributed to the migration of Spring Boot services from on-premises infrastructure to Google Cloud (GCP), moving user authentication from managed-identity validation to Microsoft Entra ID with security-group-based access control.
- Secured API-to-API communication with signed JWTs carrying scopes, so each service can only call the operations its scopes allow.
- Built an AI-powered service mapping tool that answers service verification queries in under 5 seconds instead of about 10 minutes.
- Stack: Java, SQL, Power BI, REST APIs, LLM tooling, Git.

Supply Data Analyst **Feb 2025 – Apr 2025**

ISA Interconexión Eléctrica S.A. E.S.P. | Medellín, Colombia

- Sole data engineer on the supply team: designed and ran Azure Databricks ETL pipelines over 100+ source files with 100% data availability.
- Set up CI/CD pipelines in Azure DevOps to deploy and maintain analytical solutions, replacing manual release steps.
- Built predictive models in Azure Machine Learning, reducing model execution time by 50%.
- Deployed RPA bots that automated 3 supply-chain reporting workflows for leadership.

Freelance Software Engineer **May 2024 – Jan 2025**

Self-employed | Medellín, Colombia

- Built Python REST APIs and ETL pipelines for several small clients.

Software Engineer & AI Support **May 2022 – May 2024**

Bancolombia | Medellín, Colombia

- Developed and maintained microservices that integrate the bank's chatbot with backend APIs, reducing technical debt and resolving API failures.
- Automated chatbot campaign delivery on AWS, using AWS Transfer Family (SFTP) for campaign file ingestion, reaching 1M+ customers per month.

- Helped implement CI/CD and an end-to-end SDLC in Azure DevOps with Scrum, cutting production deployment time from 48 hours to 2 hours.
- Introduced test-driven development and automated testing with Serenity BDD and Playwright, reducing production errors by 30%.
- Migrated application deployments to a cloud-native setup, eliminating per-application infrastructure cost for the migrated services.

Business Analyst Intern **Jul 2021 – Jan 2022**
Bancolombia | Medellín, Colombia

- Automated extraction of 30,000+ records from Impala (SQL) and SAS Studio into reporting workbooks, cutting manual errors by 70%.
- Designed and launched a Power BI tool, published on the bank's network, giving managers and VPs historical trends and 6-month projections.

SELECTED PROJECTS

Real Estate Analyst Colombia – Data Platform & ML
Personal project | 2026 – Present | realstateanalystcol.streamlit.app

- Orchestrated scheduled ingestion with GitHub Actions: a matrix of Playwright scrapers on hosted and self-hosted runners, resumable checkpoints, failure and zero-result alerts, and an automatic trigger of the downstream Databricks ETL job.
- Built a medallion lakehouse on Amazon S3 and Databricks with PySpark, including cross-portal deduplication, DIVIPOLA geographic normalization, listing lifecycle tracking, and unit-tested quality rules.
- Wrote Terraform for an AWS ECS Fargate deployment (ECR with lifecycle policies, task definitions, service discovery, autoscaling, CloudWatch logs, least-privilege IAM, Secrets Manager) with GitHub Actions plan/apply and image-build pipelines. Paused to keep running costs under US$1/month; the app is served from Streamlit Cloud.
- Improved the XGBoost valuation model's MAPE from 30% to 20.8% across 8 versions; predictions served through FastAPI and Streamlit.

PeraBank – Credit Risk Data Platform
Personal project | 2025 – 2026

- Built a DuckDB medallion pipeline over 2.9M LendingClub loans and four other public datasets, with schema contracts that fail the run and quality checks at every layer; 12-table snowflake model in Parquet.
- Designed a serverless AWS architecture (S3, SQS FIFO with DLQ, DynamoDB, Lambda, API Gateway, least-privilege IAM) and verified it end-to-end on a local AWS emulator with a cost model before any cloud spend; CI with Ruff and pytest on GitHub Actions.
- Trained a probability-of-default model with out-of-time validation: AUC 0.686 vs 0.675 for LendingClub's own sub-grade.

Colombian Open-Data RAG Agent – EPAM Challenge
Aug 2026

- Built a RAG agent on AWS Bedrock (Llama 3 70B) over 144,602 chunks indexed in ChromaDB from Colombian open-data APIs, with Terraform-provisioned DynamoDB usage quotas and hard token limits; 87.5% faithfulness measured with Ragas.

EDUCATION

B.Eng. in Mechatronics Engineering **Apr 2022**
Instituto Tecnológico Metropolitano (ITM) | Medellín, Colombia

Focus: Robotics and Computer Vision. Projects in ROS, deep learning object detection, and control systems.

CERTIFICATIONS

- AWS Certified Cloud Practitioner – Amazon Web Services (earned 2023; expired, renewal pending)
- DevOps Specialization: Python, Kubernetes and containers – EPAM Campus (2025)
- Advanced Artificial Intelligence Bootcamp, 159 hours – MinTIC Colombia and Universidad de Antioquia (2025)
- Applied Data Science with Python; Data Analysis with Python; Data Visualization Using Python; Data Science Math Skills

LANGUAGES

Spanish (native) | English (full professional proficiency) | French (elementary)